GRADE 9 AGRICULTURE

DETAILED FULL-YEAR LEARNER NOTES

Terms 1, 2 and 3 | Explanations • Practical activities • Worked examples • Revision

Junior School Learning Hub
Original teacher-prepared resource for learning and revision

How to use these notes

These notes are arranged around the Grade 9 Agriculture learning programme and are intended to support classroom learning, practical work and revision. Read each explanation, study the examples, complete the activities safely and attempt the questions before checking the suggested answers.

Agriculture is practical. Learners should observe real crops, soil, livestock, tools and farm records where possible. Activities involving chemicals, sharp tools, machinery, animals or unfamiliar plants must be supervised by a competent adult. Never taste unknown plants or chemicals.

Curriculum note: This is an original teacher-prepared learning resource, not an official KICD publication. Teachers should cross-check the final topic sequence, terminology and lesson allocation against the current KICD Grade 9 Agriculture Curriculum Design and the school's approved scheme of work.

Agriculture and its importance

Agriculture is the science, art and business of cultivating land, growing crops and rearing animals for food, raw materials, income and other human needs. It includes crop production, livestock production, soil and water management, farm structures, farm tools, marketing and conservation.

- Provides food such as cereals, vegetables, fruits, milk, eggs and meat.
- Provides raw materials for industries, including cotton, hides, skins and fibres.
- Creates employment in production, processing, transport and marketing.
- Generates income for households and contributes to national development.
- Supports environmental care when soil, water, biodiversity and nutrients are managed responsibly.

Key agricultural principles

Principle	Meaning in practice
Sustainability	Meet present needs while conserving resources for future generations.
Good agricultural practice	Use safe, efficient and responsible methods in production.
Biosecurity	Prevent the introduction and spread of pests and diseases.
Record keeping	Write down farm activities, inputs, outputs, sales and costs.
Value addition	Improve a product through cleaning, grading, processing, packaging or preservation.

1. Soil, soil fertility and soil conservation

Soil is the loose upper layer of the Earth's surface in which plants grow. It contains mineral particles, organic matter, water, air and living organisms. Soil supports plant roots, stores water and nutrients, and provides a habitat for organisms such as earthworms and microorganisms.

1.1 Components and properties of soil

Component	Role
Mineral particles	Sand, silt and clay form the mineral part and influence texture.
Organic matter (humus)	Decomposed plant and animal remains; improves structure and nutrient supply.
Soil water	Dissolves and transports nutrients to plant roots.
Soil air	Provides oxygen for root respiration and soil organisms.
Living organisms	Decompose organic matter, recycle nutrients and improve soil structure.

Soil texture describes the relative proportions of sand, silt and clay. Sandy soil has large particles and drains quickly but often holds fewer nutrients. Clay soil has very fine particles, holds water and nutrients strongly, but may drain slowly and become sticky. Loam contains a balanced mixture and is often suitable for many crops.

Soil structure is the arrangement of soil particles into aggregates or crumbs. Good structure allows air and water to move and roots to penetrate. Compaction reduces pore spaces and restricts root growth.

1.2 Soil fertility and plant nutrients

Soil fertility is the ability of soil to supply essential nutrients in suitable amounts and forms for healthy plant growth. Plants need macronutrients in relatively large amounts and micronutrients in small amounts.

Nutrient	Main function	Possible deficiency symptom
Nitrogen (N)	Promotes leafy vegetative growth and chlorophyll formation.	Older leaves may become pale or yellow; growth may be stunted.
Phosphorus (P)	Supports root development, energy transfer, flowering and seed formation.	Poor root growth and delayed maturity; leaves may appear dark or purplish.
Potassium (K)	Supports water regulation, enzyme activity and plant strength.	Leaf edges may yellow or scorch; plants may be weak.
Calcium (Ca)	Important for cell walls and growing points.	Growing tips and young leaves may be damaged.
Magnesium (Mg)	Central component of chlorophyll.	Yellowing between veins, often on older leaves.
Sulphur (S)	Needed for some proteins.	Young leaves may become pale.

A deficiency symptom is not proof of one nutrient shortage because pests, diseases, water stress and soil conditions can cause similar signs. Confirm a suspected deficiency through observation, soil testing or competent agricultural advice.

1.3 Maintaining and improving soil fertility

- Apply well-decomposed farmyard manure or compost to add organic matter and nutrients.
- Use suitable inorganic fertilisers according to soil test results, crop needs and label instructions.
- Practise crop rotation, including suitable legumes that can contribute nitrogen through biological fixation.
- Grow cover crops to protect the soil and add organic matter.
- Return safe crop residues to the soil where appropriate; avoid spreading diseased material.
- Use mulching to reduce evaporation, suppress weeds and protect the soil surface.

Compost is made by controlled decomposition of organic materials such as dry leaves, grass, vegetable waste and animal manure. A well-managed compost heap needs a mixture of materials, moisture, air and time. Turn the heap to introduce air and keep it moist but not waterlogged. Mature compost is dark, crumbly and has an earthy smell.

1.4 Soil erosion and conservation

Soil erosion is the removal of topsoil by water, wind or human activities. Topsoil is valuable because it contains much of the soil organic matter and nutrients needed by crops.

Type of erosion	Description / signs
Splash erosion	Raindrops detach soil particles and splash them away.
Sheet erosion	A thin, fairly uniform layer of soil is removed by flowing water.
Rill erosion	Small channels form as runoff concentrates.
Gully erosion	Large channels develop and may become difficult to cross or control.
Wind erosion	Loose dry soil is blown away, especially where vegetation cover is poor.

- Contour farming: cultivate across a slope rather than directly up and down it.
- Terracing: construct level or nearly level steps on steep slopes where suitable.
- Grass strips and cover crops: slow runoff and hold soil in place.
- Mulching: cover the soil surface with suitable plant material.
- Agroforestry: integrate trees with crops or livestock to protect soil and improve the farm environment.
- Maintain drainage channels and avoid overgrazing or unnecessary removal of vegetation.

1.5 Soil management: worked application

A farmer notices that rainwater forms channels through a bare sloping field. The likely problem is concentrated runoff and rill erosion. The farmer can establish grass strips, maintain ground cover, cultivate along contours and consider suitable terraces after obtaining technical advice.

Worked examples and applications

A farmer applies manure before planting, rotates maize with a suitable legume and mulches the crop. These practices add organic matter, improve nutrient cycling and protect the soil surface.

Practical learning activities

- Compare soil samples from a cultivated field and a vegetated area. Observe colour, texture, moisture and visible organic matter. Do not taste soil or inhale dust.

Check your understanding

1. Differentiate soil texture and soil structure.
2. State four ways of maintaining soil fertility.
3. Explain how contour farming reduces erosion.
4. Give two reasons why soil testing is useful.

Suggested answers

1. Texture is the proportion of sand, silt and clay; structure is how particles are arranged into aggregates.
2. Manure/compost, appropriate fertiliser, crop rotation with legumes, cover crops and mulching.
3. It reduces the speed of runoff by following the contour lines across the slope.
4. It identifies nutrient status and helps guide appropriate fertiliser or amendment use.

2. Crop production and management

Crop production involves preparing land, selecting suitable planting materials, establishing crops, managing them as they grow, harvesting and handling produce. Successful production depends on crop requirements, climate, soil, water, pests, diseases and good management.

2.1 Selecting crops and planting materials

A farmer should select crops suited to the local climate, soil, altitude, available water, market and household needs. Planting materials may include seeds, seedlings, tubers, bulbs, cuttings, suckers or other vegetative parts.

Selection factor	What to consider
Adaptability	Can the crop grow under the local rainfall, temperature and soil conditions?
Seed quality	Is the seed viable, clean, healthy and from a reliable source?
Maturity period	Will the crop mature within the available growing season?
Market and use	Is there demand, storage capacity and a suitable use for the produce?
Resistance/tolerance	Is the variety suitable for common local pests and diseases?

2.2 Land preparation and planting

Land preparation creates a suitable seedbed, controls existing weeds and allows roots to establish. The method depends on the crop, soil, slope, moisture and available equipment. Excessive cultivation can destroy soil structure and increase erosion.

- Clear the field selectively while retaining useful trees and ground cover.
- Remove or manage weeds before they compete strongly with the crop.
- Prepare a fine seedbed for small seeds where appropriate, but avoid pulverising wet soil.
- Mark rows and planting stations using recommended spacing.
- Plant at the recommended depth and at the onset of suitable moisture conditions.
- Label trial plots and record the planting date, variety and inputs used.

Spacing is the distance between plants within a row and between rows. Correct spacing reduces competition, allows air movement, permits field operations and can improve yield. Too close spacing may encourage disease and competition; too wide spacing may waste land.

2.3 Crop establishment and field management

- Thinning: remove excess seedlings to leave the recommended number at each station.
- Gap filling: replace missing seedlings early, using healthy seedlings of similar age where appropriate.
- Weeding: remove or control weeds before they compete strongly for light, water, nutrients and space.
- Mulching: cover soil with suitable material to conserve moisture and suppress weeds.
- Watering/irrigation: provide water according to crop stage and soil moisture; avoid waterlogging.
- Top dressing: apply fertiliser at the recommended time, rate and placement, following label instructions.
- Staking and training: support crops that need physical support and guide growth where appropriate.

Integrated pest management (IPM) combines methods to keep pests below damaging levels while reducing risks to people and the environment. It may include field monitoring, resistant varieties, crop rotation, sanitation, physical barriers, biological control and carefully selected pesticides as a last or necessary option.

2.4 Harvesting and post-harvest handling

Harvest at the correct maturity stage. Harvesting too early may reduce size, flavour or quality; harvesting too late may cause shattering, over-ripening, pest damage or spoilage. Use clean tools and containers, avoid bruising and separate damaged produce.

- Sort and grade produce according to size, quality and condition.
- Dry grains to a safe storage moisture level using appropriate guidance; damp grain can develop mould.
- Store produce in clean, dry, ventilated and pest-resistant conditions.
- Use clean packaging and keep produce away from chemicals and animals.
- Keep records of quantity harvested, losses, sales and storage conditions.

2.5 Worked crop-production example

A school garden has a rectangular bed measuring 4 m by 3 m. If a crop is planted at 0.5 m between rows and 0.25 m between plants, the approximate number of planting positions depends on the layout and whether border margins are used. Along 3 m, six row spaces may fit; along 4 m, sixteen plant spaces may fit, giving approximately 96 positions. The actual number should be planned with a clear field layout and suitable edge spacing.

Worked examples and applications

Before planting, learners should check the recommended spacing for the actual crop variety. Never assume one spacing suits all crops.

Practical learning activities

- Establish a small school garden plot. Draw a plan, label crop rows, record planting date and variety, observe germination and keep weekly growth notes.

Check your understanding

1. State five factors considered when selecting a crop.
2. Explain why correct spacing is important.
3. What is gap filling?
4. Give four post-harvest practices that reduce losses.

Suggested answers

1. Climate, soil, water, maturity period, market, household use and pest/disease tolerance.
2. It reduces competition, improves air circulation, allows field operations and supports healthy growth.
3. Replacing missing plants early with healthy planting material.
4. Sorting, grading, careful handling, proper drying, clean packaging and suitable storage.

3. Livestock production and management

Livestock production is the keeping and management of domesticated animals for products and services such as milk, meat, eggs, manure, hides, skins, income and draught power. Good management protects animal welfare, productivity and public health.

3.1 Common livestock and their products

Livestock	Examples of products / benefits
Cattle	Milk, beef, manure, hides and draught power in some systems.
Goats and sheep	Meat, milk in some systems, skins, manure and income.
Poultry	Eggs, meat, feathers and manure.
Pigs	Pork, manure and income.
Rabbits	Meat, manure and income.

3.2 Basic animal needs and welfare

- Adequate clean water and a balanced diet suitable for species, age and production stage.
- Shelter that protects animals from extreme weather and allows ventilation.
- Enough space, suitable flooring and clean resting areas.
- Regular observation for changes in appetite, movement, breathing, droppings or behaviour.
- Protection from pain, injury, unnecessary fear and avoidable disease.
- Prompt attention from a qualified animal-health professional when an animal is sick or injured.

A balanced ration supplies energy, protein, minerals, vitamins, fibre where required and water in suitable amounts. Requirements differ among species and with age, pregnancy, growth, egg laying and milk production. Mouldy feed should not be offered because it may contain harmful toxins.

3.3 Housing, hygiene and biosecurity

Livestock housing should be safe, dry, well ventilated, easy to clean and suitable for the animal species. Poor ventilation, overcrowding, wet floors and dirty feeders can increase disease risk.

- Clean housing and remove manure regularly.
- Provide clean feeders and drinkers; prevent contamination by faeces.
- Isolate new or sick animals according to veterinary advice.
- Restrict unnecessary visitors and clean footwear/equipment where appropriate.
- Dispose of carcasses and waste safely under local veterinary and environmental guidance.
- Follow vaccination, deworming and treatment advice from qualified animal-health personnel.

3.4 Signs of healthy and sick animals

Generally healthy signs	Warning signs requiring attention
Alert and responsive	Unusual dullness or isolation
Normal appetite and drinking	Loss of appetite or excessive thirst
Normal movement and posture	Lameness, weakness or difficulty standing
Clean eyes and normal breathing	Discharge, coughing or laboured breathing
Normal droppings for the species	Severe diarrhoea, blood or unusual droppings
Good body condition for age and type	Sudden weight loss or poor body condition

These are general observations, not a diagnosis. Some diseases have few early visible signs. Seek veterinary guidance rather than attempting to diagnose or medicate animals using guesswork.

3.5 Poultry management example

A small flock needs suitable housing, clean water, balanced feed, adequate space, ventilation, litter management and protection from predators. Record the number of birds, feed offered, water use, eggs collected, deaths, treatments and sales. Sudden deaths or signs of contagious disease should be reported promptly to an animal-health professional.

3.6 Livestock application

A poultry unit has wet litter and a strong ammonia smell. This suggests poor moisture control and ventilation. The keeper should improve ventilation, repair drinker leaks, remove wet litter safely and avoid overcrowding.

Worked examples and applications

A daily poultry record can show whether egg production changes after feed, water, housing or health conditions change.

Practical learning activities

- Observe animal housing in a safe setting. Identify water points, feeding points, ventilation, waste disposal and measures that protect animals from injury.

Check your understanding

1. List six basic needs of livestock.

2. Explain four biosecurity measures.

3. State five warning signs of ill health.

4. Why should animal treatment be guided by a qualified professional?

Suggested answers

1. Clean water, balanced feed, shelter, space, hygiene, health care and humane handling.

2. Cleaning equipment, controlling visitors, isolating sick/new animals, safe waste disposal and following vaccination advice.

3. Loss of appetite, dullness, lameness, abnormal breathing, discharge, diarrhoea or sudden weight loss.

4. To ensure correct diagnosis, safe medicine use, correct dosage and prevention of harmful residues or resistance.

4. Farm tools, equipment and safety

Farm tools and equipment help with land preparation, planting, weeding, irrigation, harvesting, handling and livestock care. Select tools for the task and maintain them to improve safety, efficiency and service life.

Tool / equipment	Common use	Care and safety
Jembe/hoe	Digging, weeding and making planting stations.	Keep blade secure and sharp; carry with blade facing down and away.
Panga/machete	Cutting vegetation and small stems.	Use only with supervision; keep a safe distance from others and store securely.
Rake	Collecting plant debris and levelling soil.	Inspect teeth and handle; do not leave pointed teeth exposed.
Watering can	Applying water to seedlings and small beds.	Clean after use and store upside down or in a dry place.
Wheelbarrow	Transporting soil, manure and produce.	Do not overload; check wheel, handles and balance.
Pruning shears	Cutting small branches and stems.	Keep blades clean and closed when not in use.
Sprayer	Applying approved crop-protection products.	Only trained adults should handle chemicals and spraying equipment.

4.1 Tool selection and maintenance

- Choose a tool that matches the task, crop and soil condition.
- Inspect handles, blades, joints, wheels and fasteners before use.

- Clean soil and plant material after use; dry metal parts to reduce rust.
- Sharpen cutting tools using safe procedures and adult supervision.
- Oil suitable moving metal parts lightly where recommended.
- Store tools in a dry, secure tool shed with sharp edges protected.
- Report damaged tools; do not improvise repairs that may cause injury.

4.2 Farm safety

- Wear suitable footwear and protective clothing for the task.
- Use tools only after instructions and a safety demonstration.
- Keep bystanders away from cutting, digging and spraying activities.
- Never point a blade or tool towards another person.
- Do not operate machinery or handle pesticides without appropriate training and supervision.
- Wash hands after handling soil, manure, plants or equipment.
- Keep chemicals in original labelled containers, locked away from children, food and animal feed.

4.3 Tool-use activity

When using a hoe, stand with stable footing, check that nobody is within the tool's swing area and work at a controlled pace. Stop if the handle is loose or the blade is damaged.

Worked examples and applications

A blunt tool may require more force and can slip, increasing injury risk. Maintenance is therefore both an efficiency and safety measure.

Practical learning activities

- Prepare a tool inventory. For each tool, record its name, use, condition, maintenance needed and safe storage location.

Check your understanding

1. Give four factors considered when selecting a farm tool.

2. State five maintenance practices.

3. Why must pesticides remain in their original labelled containers?

Suggested answers

1. Task, soil/crop conditions, tool size, user ability and safety.

2. Clean, dry, inspect, sharpen safely, repair properly and store securely.

3. The label carries hazard, use, storage and first-aid information; transferring chemicals can cause dangerous confusion.

5. Water management and irrigation

Water is needed for seed germination, photosynthesis, nutrient transport, cooling and plant growth. Too little water causes wilting and reduced growth; too much water can exclude air from the root zone, encourage root diseases and cause nutrient loss.

5.1 Sources and conservation of farm water

- Rainwater collected from roofs or other suitable catchments.
- Rivers, streams, dams and ponds where abstraction is permitted and safe.
- Boreholes and wells, subject to water quality and sustainable use.
- Stored water tanks and harvested runoff.
- Reuse of suitable treated water only where approved for the intended agricultural use.

Water conservation includes mulching, repairing leaks, watering at suitable times, using efficient irrigation methods, harvesting rainwater and maintaining soil organic matter. Protect water sources from manure, pesticides, detergents, fuel and other pollutants.

5.2 Irrigation methods

Method	How it works	Advantages and limitations
Watering can	Water is applied by hand to small areas.	Simple and low-cost; labour-intensive and unsuitable for large fields.
Furrow irrigation	Water flows along channels between crop rows.	Can be simple; may cause erosion or uneven distribution if poorly managed.
Sprinkler irrigation	Water is sprayed through nozzles over the crop.	Can cover larger areas; wind and evaporation may reduce efficiency.
Drip irrigation	Water is delivered slowly near plant roots through emitters.	Can save water; pipes and emitters need filtering and maintenance.

5.3 Efficient irrigation and drainage

Irrigate according to crop water needs, growth stage, soil moisture and weather. Watering early in the morning or later in the day can reduce evaporation. Avoid wetting leaves unnecessarily where this encourages disease. Drain excess water safely without causing erosion or damaging neighbouring land.

5.4 Water management application

A vegetable bed wilts during a dry spell. Check soil moisture below the surface before watering. Apply water near the root zone and mulch the bed. If water remains standing for long periods, improve drainage rather than adding more water.

Worked examples and applications

A leaking hose wastes water and may create muddy areas. Repairing it is a simple water-conservation measure.

Practical learning activities

- Compare watering-can and drip irrigation on a small plot. Record water used, time taken and plant response; do not waste water.

Check your understanding

1. State four functions of water in plants.

2. Compare sprinkler and drip irrigation.

3. Give five ways to conserve farm water.

Suggested answers

1. Germination, photosynthesis, nutrient transport, cooling and maintaining cell turgidity.

2. Sprinklers spray water over an area; drip systems deliver small amounts near roots. Drip can reduce evaporation but requires maintenance.

3. Mulching, rainwater harvesting, leak repair, efficient irrigation, watering at suitable times and improving soil organic matter.

6. Crop pests, diseases and weeds

A crop pest is an organism that damages crops or competes with them. Pests include insects, mites, rodents, birds and some other organisms. A plant disease is a harmful condition caused by pathogens such as fungi, bacteria, viruses or nematodes, or by non-living stresses. Weeds are unwanted plants growing where they are not desired.

6.1 Common pest damage

Pest group	Typical damage
Chewing insects	Holes in leaves, cut stems or damaged fruits.
Sucking insects	Sap removal, leaf curling, yellowing and sometimes transmission of viruses.
Stem/fruit borers	Tunnels or holes in stems, fruits or other plant parts.
Rodents and birds	Eating seeds, seedlings, grain or mature produce.
Nematodes	Root damage, poor growth and sometimes root galls.

6.2 Plant disease and spread

Disease development may depend on a susceptible host, a disease-causing organism and favourable environmental conditions. This relationship is sometimes called the disease triangle. Moisture, temperature, poor air circulation, infected planting materials and contaminated tools may influence disease spread.

- Use healthy planting materials and suitable resistant or tolerant varieties.
- Rotate crops to interrupt the life cycle of some pests and pathogens.
- Remove and safely dispose of infected plant parts where recommended.
- Clean tools and avoid moving contaminated soil between plots.
- Provide suitable spacing and drainage to reduce excessive humidity.
- Monitor crops regularly and seek expert identification before treatment.

6.3 Weed management

Weeds compete with crops for light, water, nutrients and space. They may also harbour pests and diseases. Control methods include hand pulling, hoeing, mulching, crop rotation, cover crops and approved herbicides applied by trained adults according to label directions.

6.4 Integrated pest management (IPM)

1. Prevent problems through crop rotation, clean seed, field hygiene and suitable varieties.
2. Monitor fields and identify the organism and level of damage.
3. Use physical, cultural or biological controls where effective.
4. Decide whether intervention is necessary based on damage and advice.
5. If a pesticide is justified, a trained adult selects an approved product and follows the label, protective equipment, waiting period and disposal instructions.
6. Evaluate results and record the action taken.

Never mix pesticides, reuse chemical containers, spray against the wind or apply products without training. Keep learners away from spraying operations and treated areas until the label's re-entry period has passed.

6.5 Pest and disease monitoring

A learner observes curled leaves and small insects beneath them. The correct first step is careful observation and identification, not immediate spraying. Record the crop, affected area, symptoms, number of plants affected and date, then seek guidance.

Worked examples and applications

Sticky traps or visual scouting may help monitor certain insects, but they do not replace correct identification.

Practical learning activities

- Make a weekly crop-health chart. Record healthy plants, symptoms, pest observations, weather and action taken. Do not collect or handle unknown harmful organisms.

Check your understanding

1. Differentiate a pest from a plant disease.

2. Explain the disease triangle.

3. Give four non-chemical pest management methods.

4. Outline the steps in IPM.

Suggested answers

1. A pest is an organism that damages crops; a disease is a harmful condition caused by pathogens or non-living stress.

2. A susceptible host, a pathogen and favourable environmental conditions interact.

3. Crop rotation, resistant varieties, sanitation, hand removal, barriers, traps and biological control.

4. Prevent, monitor, identify, choose appropriate control, use chemicals only when justified and safely, then evaluate and record.

7. Farm records and simple farm planning

Farm records are written or electronic accounts of farm activities, inputs, outputs, costs, sales, labour, livestock and stock. Accurate records help a farmer plan, monitor performance, identify losses and make informed decisions.

Record	Information recorded	Why useful
Input record	Seeds, fertiliser, feed, tools and quantities purchased or used.	Tracks use and stock.
Labour record	Date, task, hours and labour cost.	Helps estimate production costs.
Production record	Crop area, planting date, harvest quantity or animal output.	Measures productivity.
Sales record	Product, quantity, price, buyer and date.	Tracks income and marketing.
Expense record	Inputs, transport, repairs and other costs.	Supports budgeting and profit calculation.
Livestock health record	Animal identification, observation, vaccination/treatment advice and dates.	Supports health monitoring and traceability.

7.1 Simple calculations

Total cost is the sum of all expenses. Total revenue is the money received from sales. Gross profit in a simple enterprise calculation is revenue minus the stated costs. A complete farm profit calculation should include all relevant costs, including labour, transport, depreciation and other overheads where appropriate.

Measure	Formula
Total cost	Add all costs.
Total revenue	Quantity sold × selling price per unit.
Gross profit (simple)	Total revenue − stated costs.
Yield per unit area	Harvest quantity ÷ area cultivated.

7.2 Worked farm budget example

A learner's vegetable project sells 40 bunches at KSh 30 each. Revenue = 40 × 30 = KSh 1,200. If recorded expenses are KSh 750, the simple surplus is KSh 1,200 − KSh 750 = KSh 450. This is only as complete as the costs recorded; unpaid labour or other omitted costs would reduce the true profit.

7.3 Farm records activity

Prepare a weekly record for a school garden showing date, activity, inputs used, labour, observations and output. Use clear units such as kilograms, litres, hours and Kenya shillings.

Worked examples and applications

If 25 kg of tomatoes are sold at KSh 80 per kilogram, revenue = 25 × 80 = KSh 2,000.

Practical learning activities

- Create a sample record book for a garden or poultry project. Do not record personal or confidential family financial details.

Check your understanding

1. State five benefits of farm records.
2. Calculate revenue from 18 trays of eggs sold at KSh 450 each.
3. A project earns KSh 3,600 and records costs of KSh 2,850. Find the simple surplus.

Suggested answers

1. Planning, monitoring, stock control, cost calculation, evaluating productivity and decision-making.
2. 18 × 450 = KSh 8,100.
3. KSh 3,600 − KSh 2,850 = KSh 750.

8. Farm business, marketing and value addition

Agricultural marketing involves activities that move farm products from producers to consumers. It may include harvesting, sorting, grading, packaging, storage, transport, advertising, pricing and selling.

8.1 Product quality and grading

Sorting separates produce according to condition or type. Grading groups produce using agreed quality characteristics such as size, maturity, cleanliness and freedom from damage. Consistent grading helps buyers compare products and can reduce disputes.

8.2 Value addition

- Cleaning and sorting vegetables before sale.
- Drying suitable fruits, grains or herbs using hygienic methods.
- Processing milk into approved dairy products in hygienic facilities.
- Packaging produce to protect it and provide accurate product information.
- Making compost from suitable farm organic waste.
- Processing crops into flour, flour-based products or other approved foods using safe methods.

Food safety is essential. Use clean water, clean equipment, hygienic handling, suitable temperatures and safe packaging. Do not sell food that is mouldy, contaminated, spoiled or unsafe. Follow applicable public health and food handling requirements.

8.3 Choosing a market and setting a price

A farmer should consider demand, quality, season, distance to market, transport cost, storage, competition and buyer requirements. A price should cover costs and be acceptable to buyers, while recognising that market prices change.

8.4 Marketing application

A farmer has ripe tomatoes but no immediate buyer. Sorting, careful packaging and using a suitable short-term storage method may reduce damage while the farmer seeks a market. Tomatoes should not be stored in conditions that accelerate spoilage.

Worked examples and applications

A product label should be truthful. Do not claim that produce is organic, certified or chemical-free unless the claim is supported by the relevant standards and records.

Practical learning activities

- Compare two local farm products by packaging, quality, price and likely customer. Explain one possible value-addition step.

Check your understanding

1. Differentiate sorting and grading.
2. Give four examples of value addition.
3. State five factors affecting the choice of a market.

Suggested answers

1. Sorting separates by condition/type; grading groups according to defined quality characteristics.
2. Cleaning, drying, processing, packaging, preservation and compost production.
3. Demand, quality, price, season, transport, distance, storage and competition.

9. Environmental conservation and sustainable agriculture

Sustainable agriculture produces food and other products while conserving soil, water, biodiversity and the health of people and animals. Farming decisions can affect rivers, wetlands, forests, wildlife and neighbouring communities.

9.1 Environmental risks from poor farming practices

Practice or problem	Possible impact
Overgrazing	Loss of vegetation, soil compaction and erosion.
Burning crop residues	Loss of organic matter, smoke pollution and risk of uncontrolled fire.
Poor chemical handling	Poisoning, contamination of soil and water, and harm to non-target organisms.
Cultivating steep slopes without protection	Runoff, soil erosion and sedimentation of water bodies.
Poor manure storage	Runoff, odour, flies and contamination of water sources.
Removing trees and hedges	Habitat loss, reduced wind protection and increased erosion.

9.2 Conservation practices

- Agroforestry and planting suitable indigenous trees.
- Protecting riverbanks, wetlands and water catchment areas.
- Maintaining ground cover and planting cover crops.
- Composting suitable organic wastes and recycling nutrients.
- Using water efficiently and preventing pollution.
- Rotating crops and integrating crops and livestock responsibly.
- Managing grazing to avoid overstocking and land degradation.
- Reducing, reusing and safely disposing of farm waste.

9.3 Climate-smart agriculture

Climate-smart agriculture seeks to improve productivity, strengthen resilience to climate variability and change, and reduce or remove greenhouse gas emissions where possible. Examples include drought-tolerant crop varieties, water harvesting, agroforestry, soil cover, improved manure management and timely farm planning. No single practice is suitable everywhere; local conditions matter.

9.4 Conservation project

A school can establish a demonstration plot with mulch, compost, a rainwater collection system and suitable indigenous plants. Learners can compare soil moisture and plant growth over time, keeping a fair record of observations.

Worked examples and applications

A conservation activity should not plant invasive species or disturb protected habitats. Select suitable species with local expert guidance.

Practical learning activities

- Map a school or community area and identify erosion risks, water sources, waste points and opportunities for tree planting or ground cover.

Check your understanding

1. Explain sustainable agriculture.

2. Give four effects of overgrazing.

3. State five ways of protecting water sources.

4. What is climate-smart agriculture?

Suggested answers

1. Farming that meets current needs while conserving resources and environmental health for the future.

2. Vegetation loss, compaction, erosion, reduced pasture and lower productivity.

3. Keep chemicals/manure away from water, maintain buffer vegetation, dispose of waste safely, prevent runoff and use water efficiently.

4. Approaches that improve productivity, strengthen resilience to climate change and reduce emissions where possible.

10. Practical investigations, projects and assessment skills

Agriculture learning involves observing, planning, carrying out practical work, recording evidence, interpreting results and communicating findings. A good investigation changes one factor at a time where possible and keeps other conditions similar.

10.1 Planning a fair investigation

7. State a clear question or problem.
8. Make a prediction that can be tested.
9. Identify the factor to change (independent variable).
10. Identify what to measure (dependent variable).
11. Keep other relevant conditions the same (controlled variables).
12. Prepare safe materials and a method.
13. Collect observations or measurements consistently.
14. Present results in a table or graph.
15. Explain the findings and identify limitations.

10.2 Sample investigation: effect of mulch on soil moisture

Question: Does mulch help soil retain moisture? Use two similar containers with the same soil and equal amounts of water. Cover one soil surface with a measured layer of dry mulch and leave the other uncovered. Keep both in the same conditions. Observe soil moisture at the same time each day. If scales are available, weigh containers consistently. Record results and explain whether the evidence supports the prediction.

Safety and fairness: Do not use contaminated materials. Keep container size, soil amount, water volume, light and observation time as similar as possible. The investigation is a classroom model and may not represent every field condition.

10.3 Suggested Grade 9 Agriculture projects

- Establish and maintain a small vegetable plot using a crop plan and record book.
- Prepare a compost unit and monitor decomposition safely.
- Conduct a soil erosion demonstration using trays with bare soil and soil protected by vegetation or mulch.
- Prepare a simple farm budget for a selected crop enterprise.
- Design a water-saving garden irrigation plan.
- Create a livestock welfare and hygiene observation checklist using a safe visit or case study.

10.4 Practical project rubric

Criterion	Excellent (4)	Good (3)	Developing (2)	Beginning (1)
Planning	Clear goal, materials, sequence and safety plan.	Mostly clear plan with minor gaps.	Plan incomplete in some areas.	Little evidence of planning.
Practical skills	Uses methods carefully and independently where safe.	Uses methods correctly with occasional guidance.	Needs frequent guidance.	Methods unsafe or not yet understood.

Records	Accurate, dated and well-organised records.	Mostly accurate records.	Some missing or unclear records.	Few usable records.
Interpretation	Explains results using evidence and recognises limitations.	Explains most results.	Gives limited explanation.	Does not connect results to evidence.
Teamwork and responsibility	Cooperates, shares tasks and cares for resources.	Usually cooperates and acts responsibly.	Inconsistent contribution.	Needs support to participate safely.

End-of-year revision questions

1. Define agriculture and state five ways it contributes to society.
2. Distinguish soil texture from soil structure.
3. Explain the roles of nitrogen, phosphorus and potassium in plant growth.
4. Describe four methods of conserving soil on sloping land.
5. Explain why crop rotation can improve farm productivity.
6. State six factors to consider when selecting planting materials.
7. Describe five crop management practices carried out after planting.
8. Explain four ways of reducing post-harvest losses.
9. State six basic needs of livestock.
10. Give five biosecurity measures used on a livestock farm.
11. Explain why farm tools should be cleaned and stored properly.
12. Compare furrow, sprinkler and drip irrigation.
13. Differentiate pests, diseases and weeds.
14. Outline the steps involved in integrated pest management.
15. State six types of farm records and their uses.
16. A farmer sells 32 kg of beans at KSh 150 per kg. Calculate the revenue.
17. A farmer's revenue is KSh 9,500 and recorded costs are KSh 6,750. Calculate the simple surplus.
18. Differentiate sorting from grading.
19. Explain five environmental effects of poor farming practices.
20. Describe how to conduct a fair investigation on the effect of mulch on soil moisture.

Suggested revision answers

1. Agriculture is the science, art and business of growing crops and rearing animals; it provides food, raw materials, jobs, income and supports development.

2. Texture is the relative proportion of sand, silt and clay; structure is the arrangement of particles into aggregates.

3. N supports leafy growth; P supports roots, energy transfer and seed formation; K supports water regulation, enzyme activity and plant strength.

4. Contour farming, terraces where suitable, grass strips, cover crops, mulching and agroforestry.

5. It can interrupt pest/disease cycles, diversify nutrient use and include legumes that contribute nitrogen.

6. Adaptability, seed quality, maturity, market demand, intended use and resistance/tolerance.

7. Weeding, thinning, gap filling, mulching, irrigation, fertiliser application and staking/training as appropriate.

8. Harvest at maturity, sort, grade, handle carefully, dry correctly, package hygienically and store suitably.

9. Clean water, balanced feed, shelter, space, hygiene, health care and humane handling.

10. Clean equipment, control visitors, isolate sick/new animals, dispose of waste safely and follow veterinary vaccination guidance.

11. Cleaning prevents corrosion and contamination; safe storage prevents injury and extends tool life.

12. Furrow channels water between rows; sprinkler sprays water over an area; drip supplies small amounts near roots.

13. Pests are damaging organisms; diseases are harmful conditions caused by pathogens or other stresses; weeds are unwanted plants.

14. Prevent, monitor, identify, select control, use pesticides only when justified and safely, evaluate and record.

15. Input, labour, production, sales, expense and livestock health records; they support planning and monitoring.

16. 32 × 150 = KSh 4,800.

17. 9,500 − 6,750 = KSh 2,750.

18. Sorting separates by type or condition; grading groups produce using defined quality standards.

19. Erosion, water pollution, biodiversity loss, soil degradation, smoke pollution and harm to people/animals.

20. Use similar containers and soil, apply equal water, mulch one sample, keep other conditions similar, measure moisture consistently, record and compare results.

Glossary of important terms

Term	Meaning
Agroforestry	Integrating trees with crops and/or livestock on the same land.
Biosecurity	Measures used to prevent the introduction and spread of pests and diseases.
Compost	Decomposed organic material used to improve soil.
Crop rotation	Growing different crops in sequence on the same land.
Drainage	Removal of excess water from soil or land.
Fertiliser	Material applied to supply plant nutrients.
Germination	The process by which a seed begins to grow into a seedling.
Humus	Stable, dark organic matter formed from decomposed remains.
Irrigation	Artificial application of water to crops.
Mulching	Covering the soil surface with suitable material.
Organic matter	Material originating from living organisms and their remains.
Pest	An organism that causes damage or loss to crops or stored produce.
Post-harvest handling	Care of produce after harvesting to maintain quality and reduce loss.
Soil erosion	Removal of soil by water, wind or other agents.
Soil fertility	Ability of soil to supply nutrients for plant growth.
Value addition	Activities that increase usefulness, quality or market value of a product.

References and teacher review

Primary curriculum reference: Kenya Institute of Curriculum Development (KICD), Grade 9 Agriculture Curriculum Design, available through the KICD CBC Materials / Grade Nine Designs portal: https://kicd.ac.ke/cbc-materials/curriculum-designs/grade-nine-designs/

This resource is original teacher-prepared material. Before classroom use or publication, the Agriculture teacher should verify the exact strand and sub-strand sequence, lesson allocation, practical requirements and terminology against the current KICD Grade 9 Agriculture design and the school's approved scheme of work. Local agricultural recommendations should be checked with qualified extension or veterinary personnel.

End of Grade 9 Agriculture Detailed Learner Notes.